4 Power Question 2019: Elizabeth's Escape

Instructions: The power question is worth 50 points; each part's point value is given in brackets next to the part. To receive full credit, the presentation must be legible, orderly, clear, and concise. If a problem says "list" or "compute," you need not justify your answer. If a problem says "determine," "find," or "show," then you must show your work or explain your reasoning to receive full credit, although such explanations do not have to be lengthy. If a problem says "justify" or "prove," then you must prove your answer rigorously. Even if not proved, earlier numbered items may be used in solutions to later numbered items, but not vice versa. Pages submitted for credit should be NUMBERED IN CONSECUTIVE ORDER AT THE TOP OF EACH PAGE in what your team considers to be proper sequential order. PLEASE WRITE ON ONLY ONE SIDE OF THE ANSWER PAPERS. Put the TEAM NUMBER (not the team name) on the cover sheet used as the first page of the papers submitted. Do not identify the team in any other way.

Elizabeth is in an "escape room" puzzle. She is in a room with one door which is locked at the start of the puzzle. The room contains n light switches, each of which is initially off. Each minute, she must flip exactly k different light switches (to "flip" a switch means to turn it on if it is currently off, and off if it is currently on). At the end of each minute, if all of the switches are on, then the door unlocks and Elizabeth escapes from the room.

Let $E(n, k)$ be the minimum number of minutes required for Elizabeth to escape, for positive integers n, k with $k \leq n$. For example, $E(2, 1) = 2$ because Elizabeth cannot escape in one minute (there are two switches and one must be flipped every minute) but she can escape in two minutes (by flipping Switch 1 in the first minute and Switch 2 in the second minute). Define $E(n, k) = \infty$ if the puzzle is impossible to solve (that is, if it is impossible to have all switches on at the end of any minute).

For convenience, assume the n light switches are numbered 1 through n.

1. Compute the following.

- **a.** $E(6, 1)$ [1 pt]
- **b.** $E(6, 2)$ [1 pt]
- **c.** $E(7, 3)$ [1 pt]
- **d.** $E(9, 5)$ [1 pt]

2. Find the following in terms of n.

- **a.** $E(n, 2)$ for positive even integers n [2 pts]
- **b.** $E(n, 3)$ for values of n of the form $n = 3a + 2$ where a is a positive integer [2 pts]
- **c.** $E(n, n - 2)$ for $n \geq 5$ [2 pts]

3. Find an integer value of k with $1 < k < 2019$ such that $E(2019, k) = \infty$. [3 pts]

4.
- **a.** Show that if $n + k$ is even and $\frac{n}{2} < k < n$, then $E(n, k) = 3$. [3 pts]
- **b.** Show that if n is even and k is odd, then $E(n, k) = E(n, n - k)$. [3 pts]

5. Find the following.

- **a.** $E(2020, 1993)$ [3 pts]
- **b.** $E(2001, 501)$ [3 pts]

6.
- **a.** Show that if n and k are both even and $k \leq \frac{n}{2}$, then $E(n, k) = \lceil \frac{n}{k} \rceil$. [3 pts]
- **b.** Prove that if k is odd and $k \leq \frac{n}{2}$, then either $E(n, k) = \lceil \frac{n}{k} \rceil$ or $E(n, k) = \lceil \frac{n}{k} \rceil + 1$. [4 pts]

7. Find all ordered pairs (n, k) for which $E(n, k) = 3$. [3 pts]

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One might guess that in most cases, $E(n,k) \approx \frac{n}{k}$. In light of this guess, define the *inefficiency* of the ordered pair (n,k), denoted $I(n,k)$, as

$$I(n,k) = E(n,k) - \frac{n}{k}$$

if $E(n,k) \neq \infty$. If $E(n,k) = \infty$, then by convention, $I(n,k)$ is undefined.

8. **a.** Compute $I(6,3)$. [1 pt]

b. Compute $I(5,3)$. [1 pt]

c. Find positive integers n and k for which $I(n,k) = \frac{15}{8}$. [2 pts]

d. Prove that for any integer $x > 2$, there exists an ordered pair (n,k) for which $I(n,k) > x$. [3 pts]

9. Let S be the set of values of $I(n,k)$ for all n,k for which $k < \frac{n}{2}$ and $I(n,k)$ is defined. Find the least upper bound of S. Prove that your answer is correct. [4 pts]

10. Find two distinct non-integral positive rational numbers that are not the inefficiency of any ordered pair. That is, find positive rational numbers q_1 and q_2 with $q_1 \neq q_2$ such that neither q_1 nor q_2 is an integer and such that neither q_1 nor q_2 is $I(n,k)$ for any integers n and k. Prove that your answers are correct. [4 pts]

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